

LEK YONG BIN (Singapore PR)

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Aljunied, Singapore

SUMMARY

Electrical and Electronic Engineering undergraduate (CGPA: 3.92/4.00) with **Lean Six Sigma Green Belt** and product engineering internship experience at **Analog Devices**. Experienced in wafer level yield analysis, Statistical Process Control (SPC), automated test log processing, and machine learning yield prediction. Available start date: **March 2027**.

EDUCATION

Asia Pacific University of Technology and Innovation

Mar 2023 – Feb 2027

Bachelor of Electrical and Electronic Engineering with Honours

CGPA: 3.92/4.0 (Expected First Class Honours)

- Recognition of Vice-Chancellor's List for Year 1, 2 & 3
- Designed a BJT class AB amplifier and a multistage CMOS operational amplifier

WORK EXPERIENCE

Analog Devices SDN BHD, Penang

Nov 2025 - Mar 2026

Product Engineering Intern - Trim & Probe

- Performed wafer level yield and bin distribution analysis using **Statistical Process Control (SPC)** and **Population Stability Index (PSI)** to **identify parametric shifts** and reduce downstream packaging scrap cost
- Built cross platform **DLog binary parsing** and **automated decoding tools** on ETS-364, J750, HP94K, Eagle-LSI
- Developed a Wafer Quality Gate system integrating automated data pipelines and machine learning, achieving an **R² of 0.7581** in early final test yield prediction and enabling proactive identification of high-risk wafers
- Performed **correlation unit testing** for 53 semiconductor products on the ETS-364 platform

KEYSTONE CABLE (S) PTE LTD, Singapore

Dec 2024 - Jan 2025

Production Engineering Intern

- Developed an RFID based user management system, integrated Node-RED, PostgreSQL, and MQTT for centralized operator data handling
- Designed and structured relational databases to improve data consistency and traceability in production workflows
- Conducted lead time and process analysis to identify inefficiencies and improve operational flow

PERUNDING ACE SDN BHD, Johor Bahru

Jan 2024 – Feb 2024

Project Engineering Intern

- Crafted electrical and telecommunication layout using AutoCAD based on project constraint & client requirements

PROJECTS & ACHIEVEMENTS

Predictive Wafer Quality Gate for Semiconductor Packaging Using Data-Driven Decision Control

Feb 2026 - Jan 2027

- Designed a data driven system to analyse wafer probe test data and **predict final test performance** based on **ML algorithms** and **SHAP analysis** (Explainable AI) before wafers enter the packaging stage
- Applied **statistical process control (SPC)** and feature correlation analysis to accelerate **root cause analysis (RCA)**
- Implemented **GDBN** bin map pattern detection to flag abnormal wafer behaviour and improve process visibility

Micron X ESUM Case Study Competition 2026 – Top 8 Finalist

Apr 2026 - Jun 2026

International Inter Varsity Innovation Challenge 2025 – Gold Award

Aug 2025 - Oct 2025

Global Youth Innovation Design and Invention Competition 2024 – Gold Award

Jul 2024 - Aug 2024

ASBX Business Challenge 2023 – Third Prize

Jul 2023 - Dec 2023

AV Tech Competition 2023 – Bronze Prize

Oct 2023 - Nov 2023

PUBLICATIONS & RESEARCH

Research Assistant, *Malaysian Communications and Multimedia Commission Research Project*

Jul 2026 – Dec 2026

Co-author, Optimising Malaysia's Parcel Delivery Resources through a Work-Sharing Model: An Industry Readiness Assessment

Apr 2026 – Aug 2026

- Manuscript submitted to *Journal of International Logistics and Trade* (**Scopus Q2**)

Co-author, Systematic Review of Optimization Methods and AI Solutions in Hybrid Energy Systems Design and Operation

Feb 2025 – Apr 2026

- Manuscript submitted to *Archives of Computational Methods in Engineering* (**SCIE/JCR Q1**)

SKILLS

Programming	: Python, C, SQL
Data & Tools	: Pandas, Scikit-learn, Power BI, PostgreSQL, Node-RED
Engineering Tools	: MATLAB, LTspice, LabVIEW, AutoCAD, SolidWorks, DIALux, ETAP
Certifications	: Siemens PLC (S7-1200) & TIA Portal, Lean Six Sigma Green Belt
Languages	: English, Mandarin, Malay